

ECO102: Introduction to Macroeconomics

Lecture 9: Aggregate Supply and Aggregate Demand

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What is the AS-AD Model

- We have seen the relationship between the labour input and output (Real GDP) in our previous chapters
- Now, in our pursuit of looking at the economy in a different light, we will look at the ***Aggregate Supply-Aggregate Demand Model (AS-AD)***
- The AS-AD Model charts the relationship between the ***output (Real GDP)*** and the ***price level***
- This model represents the consensus view of macroeconomists on how real GDP and the price level are determined
- The model provides a framework for understanding the forces that make our economy expand, that bring inflation, and that cause business cycle fluctuations

What is the AS-AD Model

- The price level and the output level in the AS-AD model are determined similarly to how the quantity and price in a competitive market are determined
- However, there are some differences, as the AS-AD model is a model of an imaginary market for the total of all the final goods and services that make up real GDP
- Despite it being an imaginary market, the AS-AD model makes clear distinctions between **supply/demand** and **quantity supplied/demanded**

Aggregate Supply

- The ***quantity of real GDP supplied*** is the total quantity of goods and services, valued in constant base year dollars, that firms plan to produce during a given period
- This quantity depends on the quantity of labour employed, the quantity of physical and human capital, and the state of technology
- ***Aggregate supply (AS)*** is the relationship between the quantity of real GDP supplied and the price level

Aggregate Supply

- At any given time, the quantity of capital, the state of technology, and the population are fixed; only the quantity of labour changes
- The labour market can be either at full employment, above full employment, or below full employment
- At full employment, the quantity of real GDP supplied is potential GDP
- Over the business cycle, employment fluctuates around full employment, and the quantity of real GDP supplied fluctuates around potential GDP
- We usually distinguish AS between two time frames:
 - **Long-run aggregate supply**
 - **Short-run aggregate supply**

Long-Run and Short-Run Aggregate Supply

- ***Long-run aggregate supply*** is the relationship between the quantity of real GDP supplied and the price level when the money wage rate changes in step with the price level to maintain full employment
- The long-run aggregate supply curve is always vertical and is always located at potential GDP
- It is vertical because potential GDP is independent of the price level
- A 10 percent increase in the prices of goods and services is matched by a 10 percent increase in the money wage rate, causing the real wage rate to remain unchanged at its full-employment equilibrium level

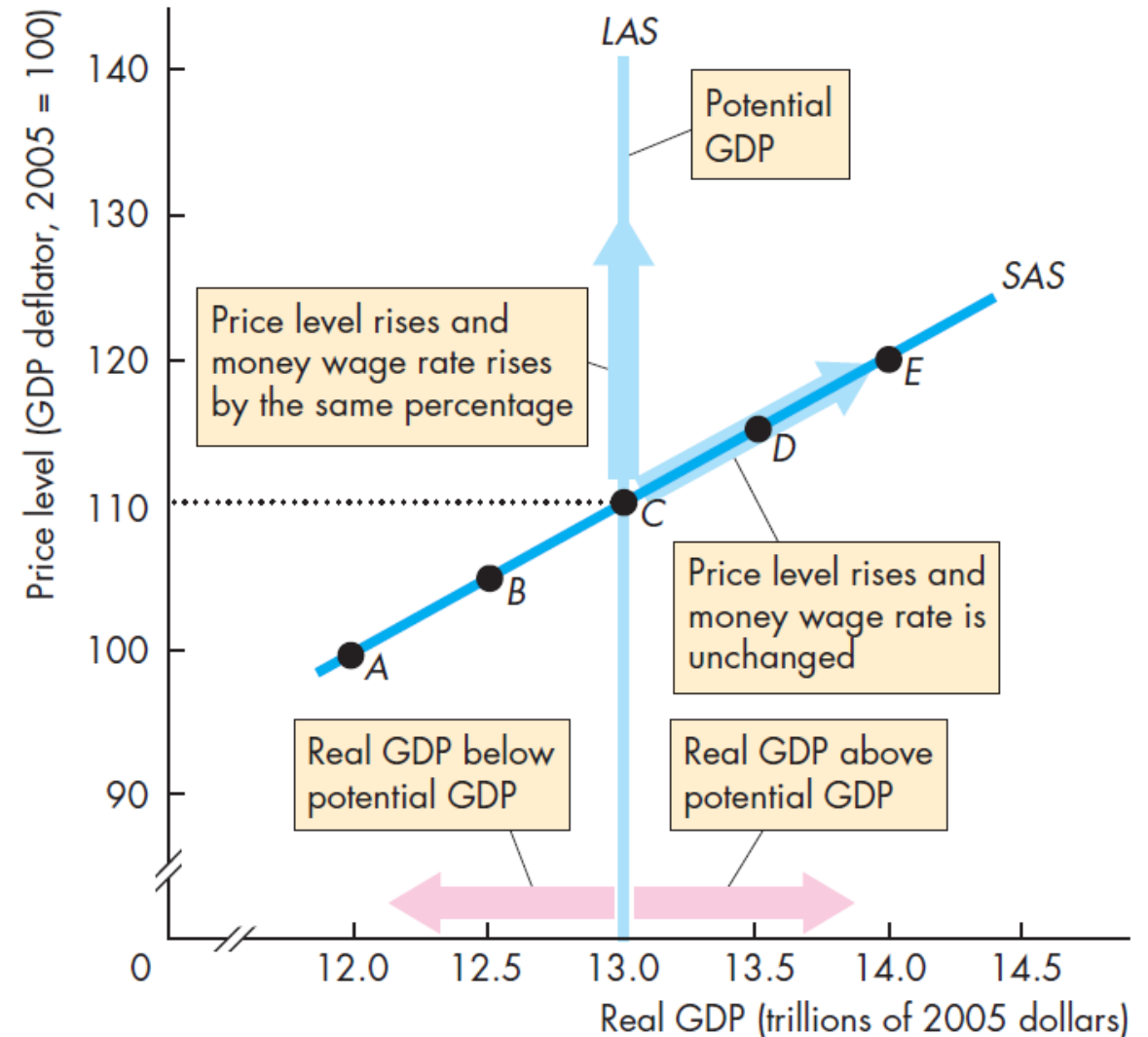
Long-Run and Short-Run Aggregate Supply

- ***Short-run aggregate supply*** is the relationship between the quantity of real GDP supplied and the price level when the money wage rate, the prices of other resources, and potential GDP remain constant
- The **Short-run aggregate supply** is determined similarly to how we have determined our supply curve for a competitive market

Long-Run and Short-Run Aggregate Supply

In the long run, the quantity of real GDP supplied is potential GDP and the LAS curve is vertical at potential GDP

In the short-run, the quantity of real GDP supplied increases if the price level rises, while all other influences on supply plans remain the same



Changes in Aggregate Supply

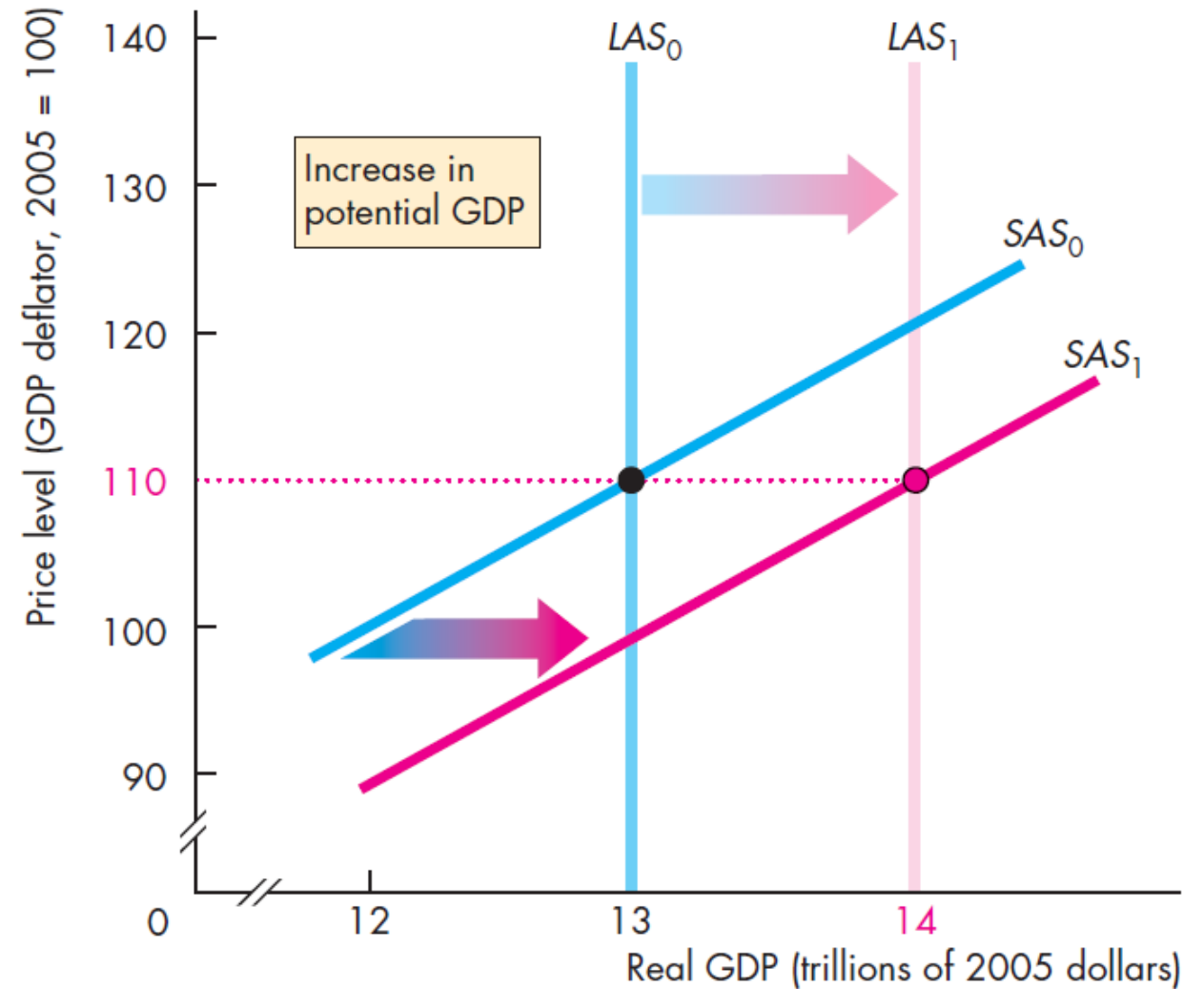
- **Aggregate supply** changes when an influence on production plans other than the price level changes
- These other influences include:
 - **Changes in potential GDP:** An increase in potential GDP increases both long-run aggregate supply and short-run aggregate supply
 - **Changes in the money wage rate:** When the money wage rate changes, short-run aggregate supply changes, but long-run aggregate supply does not change

Aggregate Supply: Changes in Potential GDP

- Potential GDP can increase for any of three reasons:
 - **An Increase in the Full-Employment Quantity of Labour:** Over time, potential GDP increases because the labour force increases; however, with constant capital and technology, potential GDP increases only if the full-employment quantity of labour increases
 - **An Increase in the Quantity of Capital:** The larger the quantity of capital, the more productive is the labour force and the greater is its potential GDP.
 - **An Advancement in Technology:** Technological advances are by far the most important source of increased production over the past two centuries

Aggregate Supply: Changes in Potential GDP

The long-run aggregate supply curve shifts rightward from LAS_0 to LAS_1 and the short-run aggregate supply curve shifts from SAS_0 to SAS_1



Aggregate Supply: Changes in the Money Wage Rate

- The money wage rate can change for two reasons:
 - **Departures from full employment:** Unemployment above the natural rate puts downward pressure on the money wage rate, and unemployment below the natural rate puts upward pressure on it
 - **Expectations about inflation:** An expected rise in the inflation rate makes the money wage rate rise faster, and an expected fall in the inflation rate slows the rate at which the money wage rate rises

Aggregate Supply: Changes in the Money Wage Rate

A rise in the money wage rate decreases short-run aggregate supply and shifts the short-run aggregate supply curve leftward from SAS_0 to SAS_2

